DSA5208 GROUP PROJECT

Rainfall across Singapore

Patterns and next-30-minute risk, 2017–2024

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Abstract

Singapore's five-minute rainfall records offer a detailed view of local weather, but gaps in the record complicate comparisons over time. We analyse the 2017–2024 annual datasets with Apache Spark, retaining 48.54 million observations from 91 station IDs after cleaning. Among sufficiently complete station-months, November has the highest mean monthly rainfall and wet intervals are most frequent in the afternoon. Comparisons between years depend strongly on which stations and months have adequate coverage. For the prediction task, we estimate the probability of exceeding a requested rainfall amount in the next 30 minutes. We compare a random forest using recent station rainfall with one using station and calendar information alone, alongside logistic-regression and historical-frequency baselines. On 5.73 million shared test cases from 2024, the recent-rainfall forest achieves a mean Brier score of 0.01613, compared with 0.02283 for the calendar forest. The calendar forest does not improve on historical frequency overall. Recent observations therefore provide useful predictive information in this experiment, while calendar-only estimates are better understood as historical risk. A subsequent comparison on 2022 validation finds little benefit from doubling the fitting sample at the reference settings, while allowing deeper trees gives A a modest gain. Supplementary gradient-boosted trees and full-training classifiers yield similar or higher probability error than the original forest; additional within-bin thresholds support the RF comparison. Large rainfall accumulations and missing observations remain important limits.

1 Introduction

The project asks two related questions: how rainfall varies across Singapore within and between years, and how well we can estimate a station's chance of receiving more than a specified amount in the next 30 minutes. The first question concerns patterns across many observations. The second requires a clear account of what information would be available when a forecast is issued.

We use all eight annual datasets from 2017 to 2024. For exploration, the main difficulty is uneven coverage: a low recorded monthly total can reflect missing observations as well as dry weather. We address this through completeness rules and comparisons at common stations. For prediction, we distinguish a calendar-only model from a model that also uses recently published rainfall. This lets us examine both the three-input query in the assignment—station, prediction time and threshold—and the benefit of supplying current weather observations internally.

The report follows these two questions from data preparation to results. Spark handles the observation-level calculations and model fitting; figures and supplementary analyses use the saved result tables.

2 Data and methods

2.1 What the records contain

We use the eight annual Historical Rainfall across Singapore datasets for 2017–2024. Each observation contains a station identifier, coordinates, a timestamp, rainfall in millimetres and update metadata. All times are interpreted in Singapore time. The measurement is an amount accumulated over a five-minute interval, rather

than a rainfall rate. The publisher cautions that records may be missing and have not undergone the quality control applied to official climate records [1].

The cleaned dataset contains 48,538,625 observations from 91 station IDs. Recorded zeros account for 97.08% of observations, while positive amounts have a long upper tail (Figure 1). We therefore examine how often rain occurs separately from how much falls during wet intervals. A zero is a measurement of no rain; a missing record provides no such information. Appendix A provides the positive-rainfall distribution and a raw-versus-clean source-gap example.

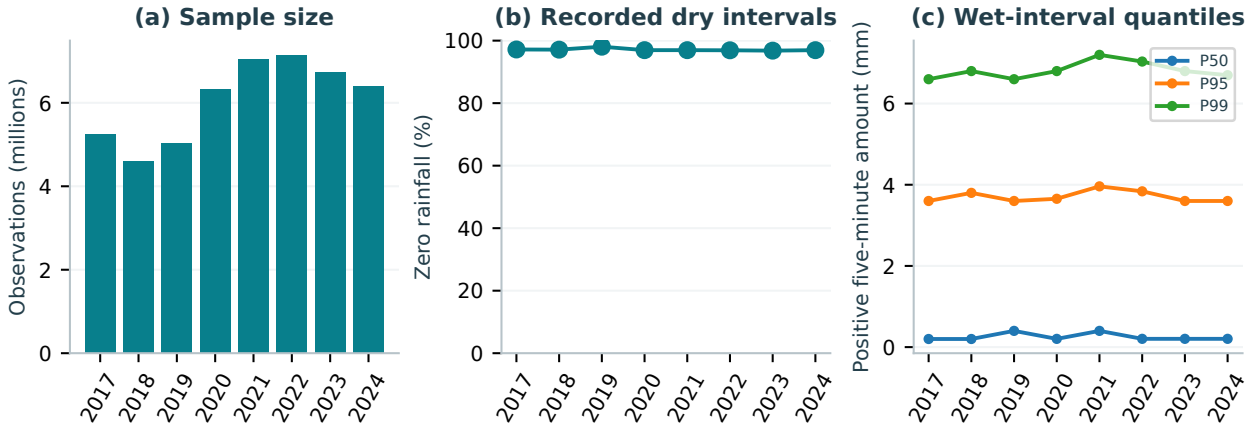


Figure 1. The observation sample by year: record counts, the percentage of recorded zeros, and quantiles of positive five-minute rainfall. Positive-rainfall quantiles exclude dry intervals.

2.2 Cleaning and missing observations

The PySpark pipeline standardises identifiers and field types, checks rainfall units and measurement type, and rejects malformed records, invalid timestamps, invalid coordinates, and non-finite or negative rainfall. Timestamps within two seconds of a five-minute boundary are rounded to that boundary; records farther from the grid are excluded. Exact duplicates are removed. If records sharing a station and time disagree on rainfall or coordinates, or include an invalid source record, all records for that station and time are set aside for inspection. Consistent measurements with different update metadata are consolidated using the latest publication timestamps, conservatively delaying their availability for prediction.

The source-to-clean audit reconciles 48,538,713 raw rows with 48,538,625 clean observations. The count falls by 88; because cleaning also consolidates repeated records, this difference is not a count of invalid measurements. All raw timestamps were parseable. High non-negative amounts are retained for review, including the maximum of 37.2 mm in five minutes; the cleaning rules alone do not establish whether an extreme reading is meteorologically valid. The pipeline preserves the raw files and writes rejected records, quality summaries and clean Parquet partitioned by year and month. The repository includes the summary evidence; the full raw and intermediate datasets require reproduction through the documented cloud workflow.

Missingness matters more to this analysis than the small change in row count. The source audit identifies 63 calendar dates without observations anywhere in the network. August 2018, for example, contains 36,410 raw records, all within its first three days. This gap was already present in the annual source file. Timestamp rounding can also move near-midnight records to an adjacent date, so daily raw-to-clean differences need not represent deletions.

For station s, year y and month m, we measure completeness as

$$C_{sym} = \frac{N_{sym}}{288 D_{ym}},$$

where N_{sym} is the number of valid unique observations and D_{ym} is the number of calendar days. Every station is represented in all 96 year-months; absent months have zero observations and an undefined rainfall total. Low coverage can consequently reflect a station's operating period as well as missing records.

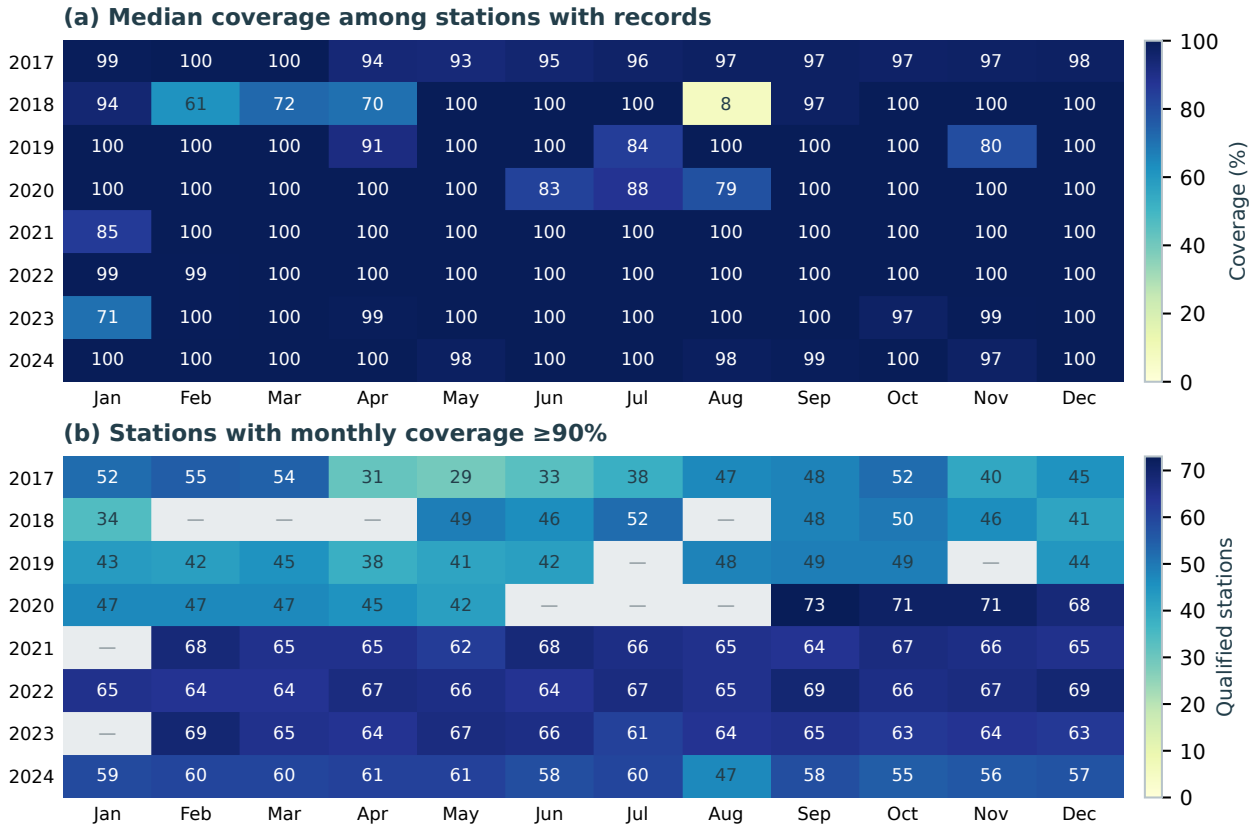


Figure 2. Monthly data availability. The upper panel gives median coverage among stations with any observations; the lower panel counts stations meeting the 90% rule. Grey cells in the lower panel have no qualifying stations.

2.3 Comparing rainfall over time

Monthly summaries require at least 90% coverage. Annual summaries require 95% coverage overall and 90% in every month, preventing a concentrated seasonal gap from being hidden by a high annual average. Daily-distribution summaries require 95% daily coverage within eligible months. These rules limit missing-data exposure but cannot remove bias if missingness is related to weather. Totals remain observed amounts, without gap filling or scaling by inverse coverage.

Within each year-month, we average eligible station totals equally. Seasonal curves then weight available years equally. Wet frequency is the number of positive intervals divided by observed intervals; conditional intensity is their rainfall sum divided by positive intervals. Hourly summaries pool eligible observations, giving more weight to periods with more observed intervals. Station means describe the sampled network, rather than an area-weighted national rainfall total.

Cross-year comparisons match station IDs within the same calendar month for each year pair. Coordinate stability screens use a tolerance of 0.005 degrees per axis within and between paired periods. We average the within-station changes and express percentage change relative to the earlier pair mean. Plots show only comparisons with at least five common stations; this is a display threshold, not a significance test. Different year pairs can contain different stations.

2.4 Computing environment

Observation-level processing, feature construction and Spark MLlib training ran on Google Cloud Dataproc. Table 1 records the experimental configuration. Only saved summaries were downloaded for local analysis and plotting; source snapshots and job metadata accompany the results.

Component	Configuration
Cluster	One master and two workers in `asia-southeast1-c`
Each VM	`e2-standard-4`: 4 vCPUs, 16 GB RAM, 80 GB `pd-balanced` boot disk
Worker capacity	8 vCPUs and 32 GB RAM, excluding the master
Runtime	Dataproc 2.2.87, Debian 12; Spark/PySpark 3.5.3 on YARN
Spark settings	6 GB driver memory; 1 GB maximum driver result; 128 shuffle partitions; adaptive SQL; Singapore timezone
Local figures	Apple M2, 16 GB RAM; pandas, Matplotlib and Plotly
Storage	Cloud Storage for raw files, Parquet data and result tables

Table 1. Configuration used for the reported experiments.

3 Rainfall patterns, 2017–2024

3.1 Seasonality and changes between years

The clearest seasonal feature is the increase in rainfall towards the end of the year. November has the highest mean observed monthly amount, 320.8 mm, while February has the lowest, 146.1 mm. These averages give equal weight to each available year after averaging its qualified stations. Figure 3 shows the individual year-months behind this summary; Figure 4 separates rainfall amount into frequency and intensity. The broad pattern agrees with the wetter November–January period and relatively dry February described by the Meteorological Service Singapore [2].

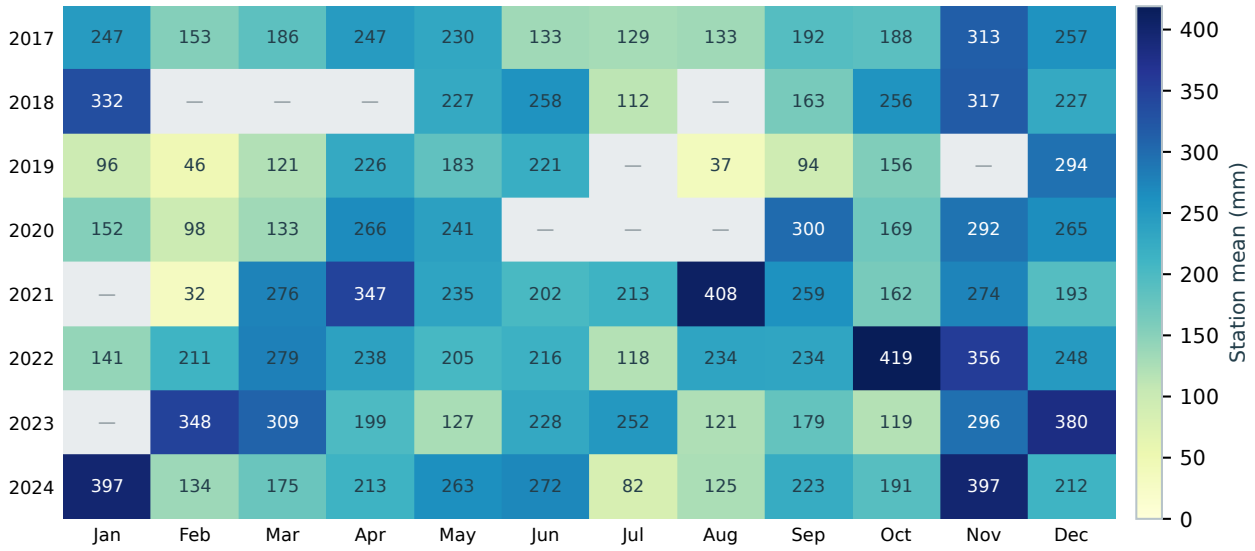


Figure 3. Mean observed monthly rainfall at qualified stations, 2017–2024. All cells share one colour scale. Grey cells have no qualified station-months.

The frequency and intensity curves help explain what makes a month wet. November has the highest mean wet-interval frequency, approximately 4.30%, but its mean amount per wet interval is about 0.89 mm. March has a higher conditional intensity, about 1.06 mm, despite a lower monthly total. Thus, November's high accumulation is associated with more frequent rain, rather than the largest amount in each wet interval. This distinction matters for prediction: a season with frequent rain need not have the greatest risk of a large amount within a short window.

Monthly totals also reflect the number of days in a month. February's shorter duration contributes to its low total, so the amount curve alone cannot establish that it has the lowest propensity for rain. The frequency and intensity panels provide complementary measures. Between six and eight years contribute to each calendar-month average, and the shaded ranges describe their observed variation rather than uncertainty in a long-term climate normal.

To examine changes between years, we compare the same calendar month at stations present in both periods

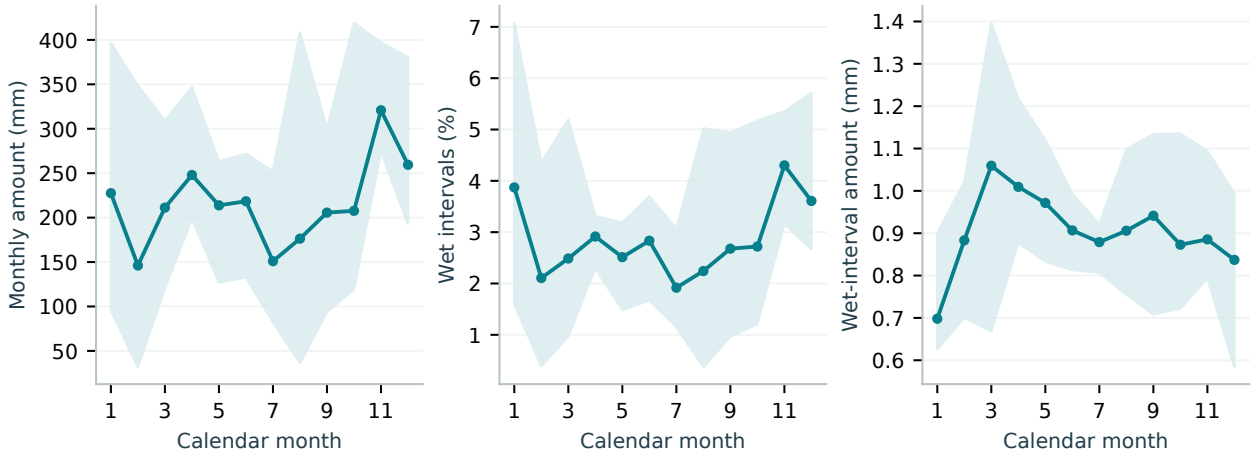


Figure 4. Seasonal means of monthly rainfall, wet-interval frequency and rainfall per wet interval. Available years receive equal weight; shading shows their minimum–maximum range.

and meeting the location-stability checks. December 2024, for example, had a mean of 213.9 mm across 53 common stations, compared with 379.8 mm in December 2023: a reduction of 43.7%. This provides stronger evidence of a change at those locations than a comparison of two unmatched network means. Figure 5 applies the same approach to adjacent-year monthly pairs. The participating stations vary between pairs, so the percentages describe separate comparisons and cannot be chained into a national trend.

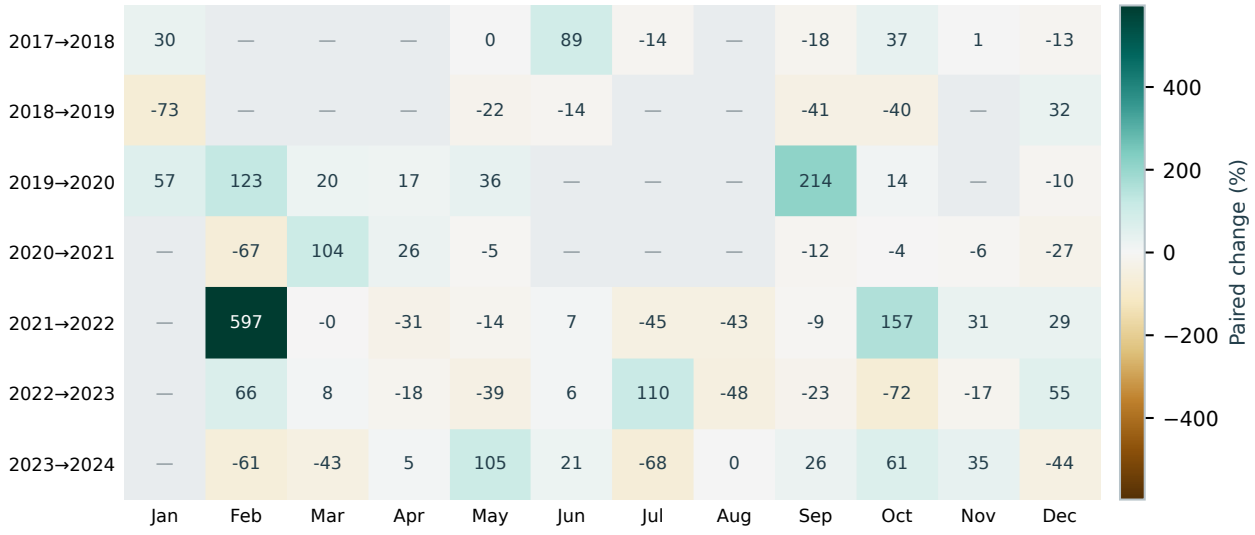


Figure 5. Adjacent-year changes in monthly rainfall at common stable-location stations, relative to the earlier year's mean. Grey cells lack support from at least five common stations; this cutoff is a display rule, not a significance test.

Full-year comparisons are more limited. Only 2017, 2022 and 2024 contain stations meeting both the annual and every-month completeness requirements. Their mean observed annual totals are 2,396.6 mm (9 stations), 2,920.9 mm (54 stations) and 2,682.3 mm (29 stations), respectively. Among the 26 stations eligible in both 2022 and 2024, the mean declined by 284.8 mm, or 9.6%. Comparisons involving 2017 have only three or four common stations. These results support a limited comparison of monitored locations, but neither an eight-year annual trend nor a national wettest-year ranking. Months with adequate coverage in the other years remain part of the monthly analysis.

3.2 When and where rain occurs

Rain is more frequent in the afternoon. Across valid intervals in qualified station-months, wet frequency reaches 5.80% at 15:00–15:59 and falls to 1.10% at 23:00–23:59, using Singapore time. Figure 6 shows how this daily pattern varies by month and year. The afternoon enhancement is consistent with MSS's description

of Singapore's rainfall [2] and motivates including hour-of-day features in the predictive model. These pooled frequencies describe observed occurrence; they are not forecasts for a particular afternoon, and the yearly curves can contain different mixes of months and stations.

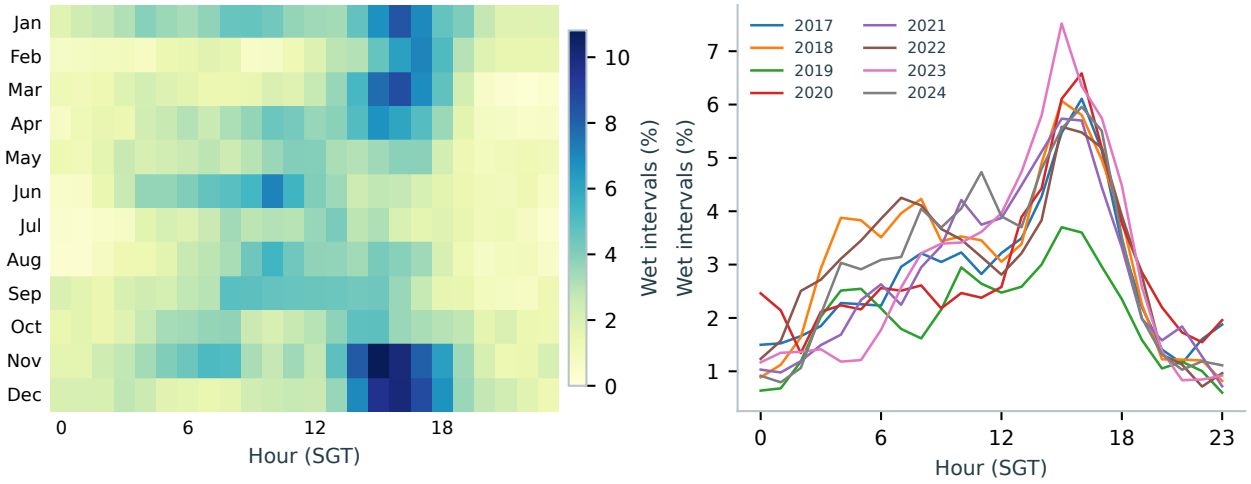


Figure 6. Wet-observation frequency by month and hour, and by year and hour, in Singapore time. Frequencies are weighted by the number of observed intervals.

Rainfall also varies substantially across monitoring locations. In December 2024, qualified station totals range from 29.5 mm at S224 to 326.8 mm at S117. The December maps retain 41–69 stations in each of the eight years, making this month a useful illustration of spatial and year-to-year variation. Figure 7 shows the measured station values on a common scale. The full set of 96 year-month maps remains in the project repository's supplementary gallery. The map outline comes from the retained boundary assets [3]. We avoid interpolating a continuous surface because the station network alone does not establish how rainfall varies between gauges; station means likewise should not be interpreted as area-weighted national rainfall.

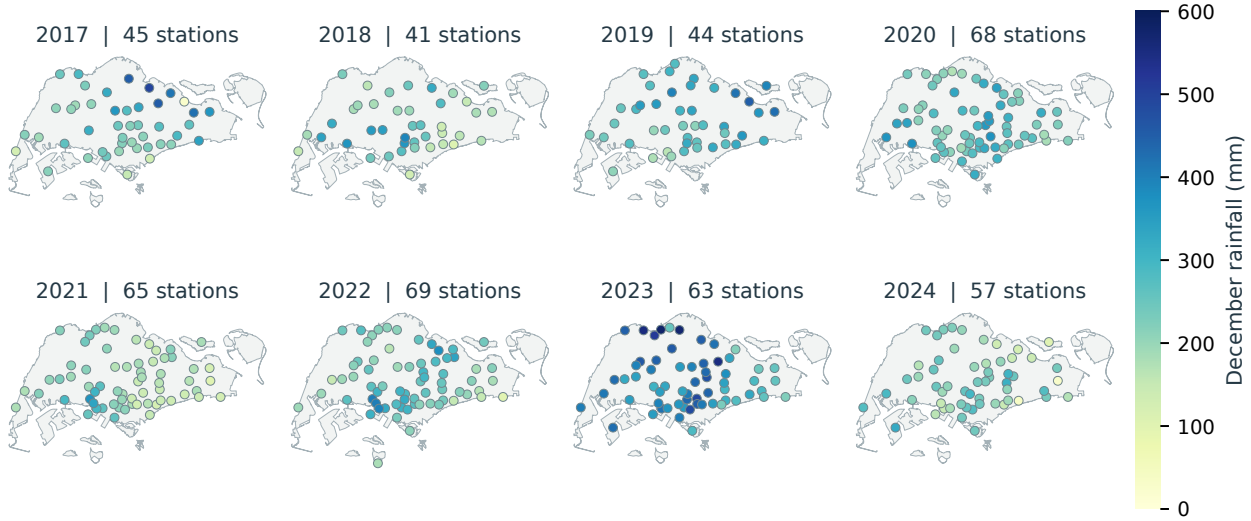


Figure 7. December rainfall at qualified stations in each study year. All eight panels use a 0–600 mm scale, covering the full range of December station totals. Panel labels give station counts; colours represent measurements at gauges, with no spatial interpolation.

3.3 How sensitive are these summaries to coverage?

Changing the monthly coverage cutoff produces relatively small changes in the supported monthly means. Relative to the 90% cutoff, using 80% gives a median absolute change of 0.61% across 85 shared year-months, with a maximum of 5.78%. Using 95% gives a median change of 0.40% across 84 shared year-months, with

a maximum of 2.97%. This is reassuring for the descriptive amount summaries, although each cutoff changes the contributing stations. It does not establish that missing observations are random. Residual gaps, uneven station coverage and the short study period remain the main reasons to treat these findings as patterns in the available observations rather than evidence of long-term climate change.

Appendices B–D retain monthly trajectories, annual station totals, daily-rainfall summaries, coverage-sensitivity curves and additional spatial views. They supplement the principal comparisons without requiring all stations to qualify in all eight years.

4 Predicting the next 30 minutes

4.1 What information is available when a prediction is made?

For station s, issue time t and threshold q in millimetres, the target is

$$Y_{s,t} = \sum_{j=1}^{6} r_{s,t+5j\,\text{min}}, \qquad p_{s,t}(q) = P(Y_{s,t} > q \mid X_{s,t}).$$

We assume that observation timestamps mark the endpoints of five-minute accumulation intervals. A training label requires all six exact future records; taking the next six available rows could incorrectly span a gap. Totals are rounded to 0.001 mm, and exceedance means strictly greater than q. Eligibility depends on these local windows, rather than Task 1's monthly coverage filter.

We interpret the assignment's station, prediction time and rainfall threshold inputs under two information settings. In A, recent station observations are available when the forecast is issued and are retrieved internally. In B, only station and calendar information are available, so the output describes historical risk for the requested time. We use A as the primary short-term forecasting model and B as a supplementary calendar-only baseline. Comparing them assesses the practical benefit of recent observations, although their different training populations prevent a controlled feature-ablation interpretation.

A, the short-term model, also reads recent rainfall from a supplied clean observation table. We use a prespecified ten-minute buffer, with twelve observations ending at $t-65, t-60, \ldots, t-10$ minutes. Each record must be present and both recorded update timestamps must be no later than t. Missing or not-yet-updated history excludes an example instead of becoming zero rainfall.

Table 2 shows a discrepancy between observation and update times, measured as the later of the two update timestamps minus the aligned observation timestamp. Median delays are close to ten minutes in both periods. In practice, the latest rainfall observation may therefore be unavailable when a forecast is issued. We use a ten-minute buffer to represent this reporting lag, together with the record-level update checks described above, rather than assuming that measurements are available immediately.

Period	Observations	Median (min)	P90 (min)	Within 10 min (%)
2017–2021 (training)	28,256,245	9.58	15.08	68.93
2017–2024 (all years)	48,538,625	9.47	12.68	76.40

Table 2. Observation-to-update delay in the cleaned records. All observations have both update timestamps. P90 is the 90th percentile; the final column is the percentage with the later update no more than ten minutes after observation. These counts describe observations, not eligible prediction examples.

Figure 8 shows the separation between predictors and the future label.

B, the calendar model, needs only station, time and threshold. It estimates the historical risk associated with that place and time of year or day. For example, B can answer a query about tomorrow afternoon; A can use tomorrow's recent observations only when they become available. Neither model uses neighbouring stations, radar or numerical weather forecasts.

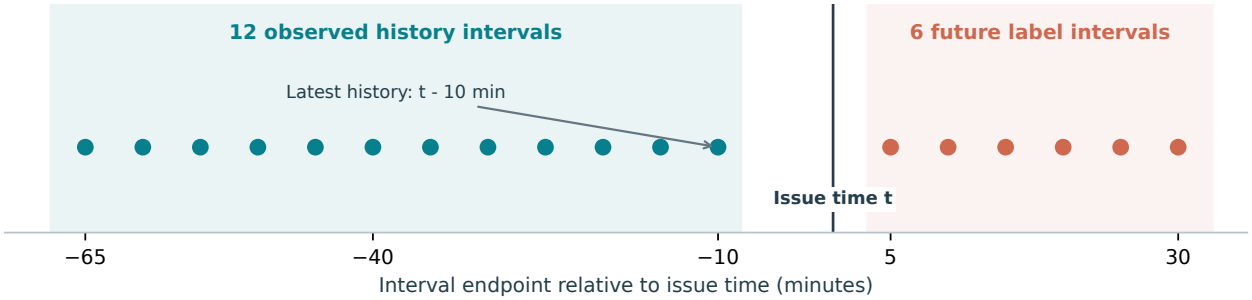


Figure 8. The information used by A at issue time t. Markers show interval endpoints. The latest history interval ends ten minutes before issue time; the six intervals after issue time form the target.

Both models include station ID, latitude, longitude, and sine/cosine encodings of time of day and annual phase. The annual encoding accounts for leap years; numeric year is excluded. A adds the latest three five-minute amounts, 15-, 30- and 60-minute totals, the 30-minute maximum, the hour's wet count and dry fraction, and the difference between consecutive 15-minute totals. The rainfall summaries represent recent intensity, persistence and change. Cyclic time features capture the seasonal and daily patterns without introducing a discontinuity at midnight or New Year. This gives A sixteen numeric features and B six, each alongside station ID.

4.2 Training and model selection

Adjacent examples share much of their history and future rainfall, so a random train/test split would mix closely related observations. We instead assign separate years to training, selection, calibration and testing, removing labels whose target end reaches the following split boundary.

Period	A examples	B examples	Purpose
2017–2021 (full)	18,688,263	27,434,001	Empirical supports and baseline
2017–2021 (15%)	2,802,737	4,115,222	RF and binary fitting
2022	6,133,684	7,107,898	Model and cutoff selection
2023	5,465,591	6,719,274	RF temperature selection
2024	5,729,508	6,374,476	Final held-out evaluation

Table 3. Eligible station-time examples after completeness checks and boundary purging. Every eligible example is used in validation, calibration and testing.

A has fewer examples because it additionally requires complete, published history. For classifier fitting we take a deterministic, label-independent 15% sample using station, timestamp and seed 2026. This initially reduced fitting work. Section 6 examines the effect of increasing the sample to 30% and broadening the forest settings; the 2024 results below remain those of the original models. Above 20 mm, the samples contain 2,886 examples for A and 4,198 for B, including overlapping windows from the same rainfall episodes. The empirical distributions and historical baseline use all eligible training examples.

Our main Spark MLlib model is a random forest that assigns the future total to seven bins: $\{0\}$, $(0, 1]$, $(1, 2]$, $(2, 5]$, $(5, 10]$, $(10, 20]$ and $(20, \infty)$ [4]. Rather than train a classifier for every possible query, we combine its bin probabilities with the observed distribution of amounts inside each bin:

$$\widehat{p}(q \mid X) = \sum_{k=0}^{6} \widetilde{\pi}_k(X)\widehat{S}_k(q), \qquad \widetilde{\pi}_k = \frac{\pi_k^{1/T}}{\sum_\ell \pi_\ell^{1/T}}.$$

Here $\pi_k(X)$ is the forest probability for bin k, and $\widehat{S}_k(q)$ is the fraction of training amounts in that bin exceeding q. The temperature T adjusts how concentrated the bin probabilities are; $T = 1$ leaves them unchanged. The resulting probability cannot increase when the rainfall threshold increases. However, the within-bin distribution is shared across all weather conditions, and its upper support ends at the largest training total, 71.8 mm. Returning zero at or above that value reflects the model's support, not physical impossibility.

The training pipeline indexes station IDs as categorical features and fits numeric median imputation; this does not relax A's complete-history requirement. We compare depths 5 and 8, holding 40 trees, 25 minimum examples per node, square-root feature subsets and 0.8 subsampling fixed. Selection minimises the equally weighted mean Brier score over $q \in \{0, 0.2, 1, 2, 5, 10, 20\}$ mm on 2022. Both models select depth 8. On 2023, both select $T = 1$ from $\{0.5, 0.75, 1, 1.25, 1.5, 2\}$, leaving their probabilities unchanged. Evaluation covers these seven thresholds; accepting other numeric thresholds does not establish their accuracy.

Two baselines help establish what the forest adds. Logistic regression uses the same features plus the query threshold, with each sampled example expanded to the seven thresholds. Station IDs are one-hot encoded and numeric inputs scaled; L2 regularisation 0.01 beats 0.1 on validation for both modes. These probabilities have no separate calibration fit. The historical baseline instead estimates station/month/hour event frequencies, shrunk towards the training global rate with 100 pseudo-observations; unseen groups use that global rate. It is a meaningful competitor for B because it also captures seasonal and daily patterns.

5 Prediction results

5.1 Does recent rainfall improve the forecast?

Our main score is the Brier score,

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N}(\hat{p}_i - z_i)^2, \qquad z_i = \mathbf{1}(Y_i > q),$$

which measures probability squared error; lower is better. PR-AUC summarises the precision–recall tradeoff, while ROC-AUC measures ranking across true- and false-positive rates. We also report log loss, which penalises confident errors, and reliability, which compares predicted probabilities with observed frequencies. Spark computes AUC using 1,000-bin curve downsampling, and its PR-AUC is not average precision [5].

For the direct comparison, both forests predict the same 5,729,508 examples, covering 89.9% of B's eligible 2024 set. Their labels agree. Training populations still differ, so this compares the fitted systems rather than isolating the effect of history features on identical training rows.

q (mm)	Event rate (%)	Brier A	Brier B	PR-AUC A	PR-AUC B
0	6.398	0.03798	0.05924	0.6091	0.1097
0.2	4.892	0.03090	0.04614	0.5582	0.0881
1	2.579	0.01870	0.02499	0.4215	0.0484
2	1.686	0.01349	0.01652	0.3155	0.0329
5	0.829	0.00732	0.00820	0.1885	0.0176
10	0.378	0.00354	0.00376	0.1020	0.0085
20	0.100	0.00098	0.00100	0.0304	0.0029

Table 4. Forest performance on identical 2024 examples. The event rate gives context for rare-event scores: small Brier values at large thresholds alone do not establish useful prediction.

Across the seven thresholds, mean Brier falls from 0.02283 for B to 0.01613 for A, a 29.4% relative reduction in probability squared error. The advantage is greatest for any positive rain (35.9% reduction) and decreases to 1.7% above 20 mm. Recent rain is therefore useful in this holdout, but the heaviest events remain difficult.

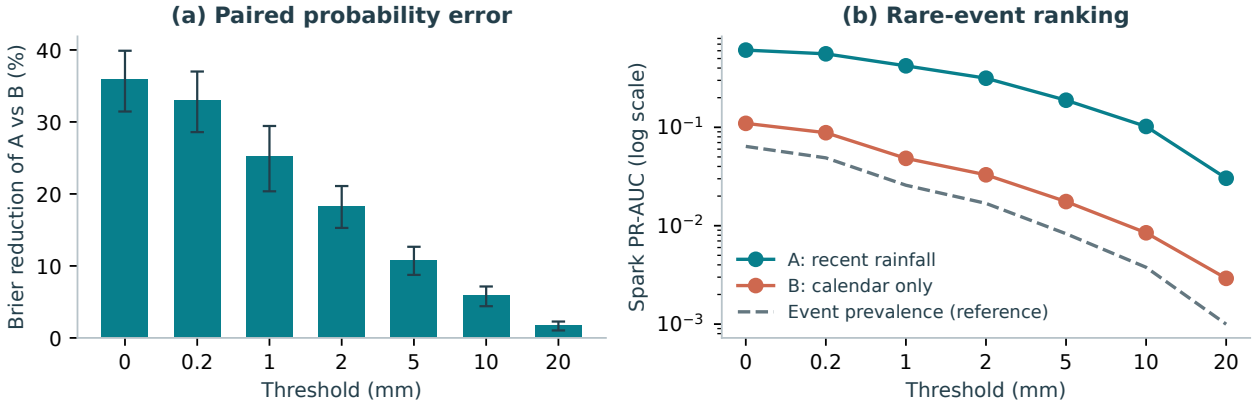


Figure 9. Comparison on shared 2024 examples. Left: relative Brier reduction with conditional 95% block-bootstrap intervals. Right: PR-AUC and event prevalence on a logarithmic scale. Rainfall thresholds are displayed as equally spaced categories.

To account for overlapping labels and shared weather across stations, we form 2,000 bootstrap replicates by resampling circular blocks of seven consecutive calendar days, retaining all stations and thresholds together. The resulting 95% interval for the mean relative Brier reduction is 25.4–32.9%. These results show that Model A consistently achieves lower Brier scores than Model B across the seven evaluated thresholds on the shared 2024 test set.

5.2 What the baselines and probability checks reveal

Mode	Model	Mean Brier	PR-AUC at 1 mm	Log loss at 1 mm
A	Random forest	0.01613	0.4218	0.0824
A	Binary LR	0.01944	0.3629	0.0875
A	Historical frequency	0.02270	0.0467	0.1166
B	Random forest	0.02274	0.0494	0.1165
B	Binary LR	0.02301	0.0428	0.1165
B	Historical frequency	0.02252	0.0516	0.1146

Table 5. Baselines on each mode's eligible test population: 5,729,508 examples for A and 6,374,476 for B. Use Table 4 for direct A/B comparisons. Small repeated AUC differences arise from Spark's partition-dependent curve downsampling.

A's forest improves Brier over both baselines at every evaluated threshold. Logistic regression nevertheless has higher ROC-AUC at 1 mm (0.8607 versus 0.8035), despite worse PR-AUC, Brier and log loss. For B, historical frequencies achieve the lowest mean Brier. More complex modelling has not demonstrated an overall benefit when only station and calendar information are available.

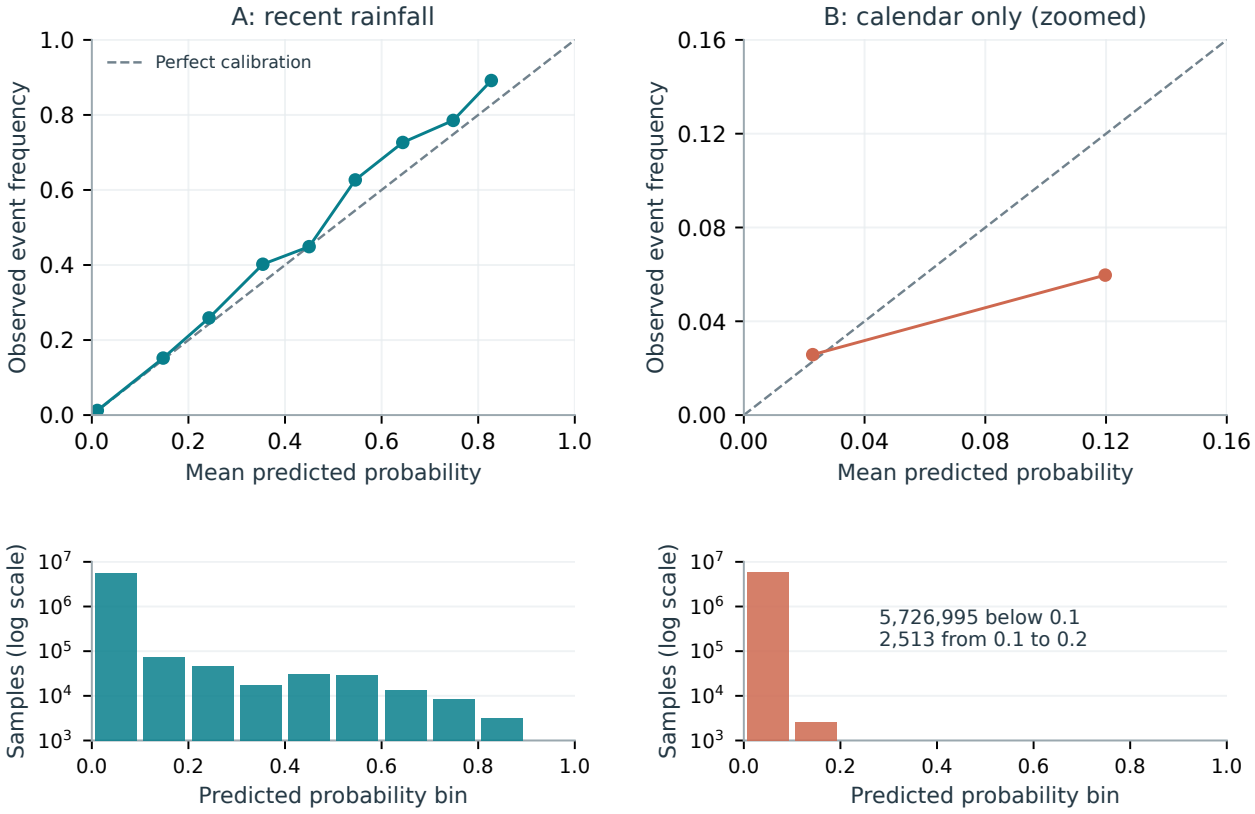


Figure 10. Calibration at 1 mm on the shared examples. Upper panels compare bin-average predictions with observed event rates; B is enlarged to show its two occupied bins. Lower panels show sample counts in fixed 0.1-wide probability bins across the full probability range.

A distinguishes a wider range of risks, but several high-probability bins underestimate the event frequency. Mean predicted risk is 2.38%, against 2.58% observed, and the ten-bin weighted absolute calibration error is 0.196 percentage points. These figures use A's full eligible set, which is also the shared set. The small aggregate error needs to be read alongside the uneven bin counts; it cannot establish reliable probabilities at every risk level.

Finally, a rainfall threshold and an alarm cutoff are different choices. At $q = 1$ mm, changing A's probability cutoff from 0.5 to the validation-selected 0.2 raises recall from 24.7% to 46.4%, while precision falls from

69.3% to 47.4%. Cutoffs are selected by 2022 F1 after fixing the 2023 temperature, without using 2024 outcomes. An always-negative rule already achieves about 97.4% accuracy here, which is why probability quality and the precision–recall tradeoff are more informative than accuracy.

Appendix E provides threshold-specific baseline skill, monthly skill and feature-importance diagnostics for the original evaluated models.

6 Does a larger training sample improve the forest?

The original 15% training sample was chosen to limit computation. To assess what this choice cost in predictive performance, we compared it with a 30% sample while keeping the reference random-forest settings fixed: 40 trees, maximum depth 8, a minimum of 25 observations per leaf, square-root feature selection and subsampling of 0.8. Both samples use the same deterministic station–time hash and seed 2026, so every anchor in the 15% sample also appears in the 30% sample. The larger samples contain 5,608,769 examples for A and 8,233,882 for B. Within each information setting, every candidate is evaluated on the same complete set of 2022 validation anchors. The training-derived distributions within rainfall bins remain fixed, and temperature is set to 1 throughout. This separates the effect of additional fitting data from changes in calibration or evaluation samples.

We also tested whether the reference forest's configuration constrained its performance. At 15%, we changed one setting at a time: depth 8 to 10, 40 to 80 trees, minimum leaf size 25 to 50, or square-root feature selection to half the features. The configuration with the lowest mean validation Brier across the seven thresholds was then fitted on the 30% sample. These comparisons broaden the earlier depth-5/8 search, but do not explore every combination of settings. Table 6 presents both information settings together.

Configuration	Fitting sample	A mean Brier	B mean Brier
Reference	15%	0.01669750	0.02332733
Reference	30%	0.01669544	0.02332800
Depth 10	15%	0.01664745	0.02331832
80 trees	15%	0.01669631	0.02332818
Minimum leaf size 50	15%	0.01671066	0.02333420
Half-feature selection	15%	0.01668206	0.02331674
Selected setting	30%	0.01663765	0.02330797

Table 6. Mean Brier on the full 2022 validation set, averaged over seven rainfall thresholds. Lower is better. Changes at 15% vary one setting from the reference. The selected 30% row uses depth 10 for A and half-feature selection for B; it is not the same configuration for both modes.

At the reference settings, doubling the fitting sample made little difference to mean Brier. For A, the score fell from 0.01669750 to 0.01669544, a relative improvement of 0.012%; for B, it rose slightly from 0.02332733 to 0.02332800, a relative deterioration of 0.003%. The conditional 95% intervals for relative improvement were $[-0.014\%,\ 0.039\%]$ and $[-0.070\%,\ 0.047\%]$, respectively. These results support using 15% as a computational compromise for the reference forests. They do not establish that the two sample sizes are equivalent or that further data could never help.

For A, increasing depth was more useful than simply increasing the sample. Depth 10 at 15% reduced mean Brier by 0.300% relative to the reference forest, with a conditional interval of $[0.213\%,\ 0.389\%]$. Increasing this depth-10 sample to 30% produced a further 0.059% improvement, with an interval of $[0.036\%,\ 0.082\%]$. The resulting score, 0.01663765, was the lowest among the tested A configurations. For B, half-feature selection gave the lowest score at 15%; its 30% version reached 0.02330797. However, the intervals for both the initial configuration change and the subsequent sample increase included zero. The evidence for a validation gain is therefore less clear for B. All of these improvements are modest in magnitude. At the reference settings, A's Brier improves by 0.103% at 10 mm and 0.072% at 20 mm, while B's changes at these thresholds are negligible and slightly adverse. The follow-up records Brier scores; it does not establish whether sample size changes PR-AUC or calibration.

The intervals use 2,000 paired circular seven-day block resamples, keeping stations and thresholds together to account for temporal dependence. They condition on the fitted models and one seed; they exclude refitting uncertainty and are not adjusted for selecting a configuration. This follow-up uses only 2017–2021 training and 2022 validation data. The candidates may benefit from selection on that validation year and have no new independent test result. The original calibrated models and their reported 2024 results remain unchanged.

7 Supplementary GBT and full-training experiments

7.1 Threshold-conditioned gradient boosting

The group also tested Spark MLlib gradient-boosted trees (GBT) with 15%, 30% and 100% of the eligible training examples. This is a binary classifier of $\mathbf{1}(Y > q)$: the rainfall threshold is an input feature alongside the corresponding A or B features. Each training anchor is expanded to the seven original query thresholds. In contrast, the main random forest predicts seven rainfall classes and combines them with training-derived within-bin distributions.

The GBT pipeline indexes station IDs, imputes numeric features and assembles them with q. It uses 40 boosting iterations, step size 0.1, minimum leaf size 25, subsampling 0.8, maximum feature bins 128 and seed 2026. Depths 3 and 5 are compared on 2022 mean Brier; temperature is selected on 2023, and results are exported for 2024. All reported configurations select depth 5; A selects temperature 0.75 and B selects 1. The imported 30% and 100% runs fit 5,608,769 and 18,688,263 anchors for A, and 8,233,882 and 27,434,001 for B. Their test counts are 5,729,508 and 6,374,476 per threshold, respectively.

Mode	Training	Mean Brier	Mean log loss	Mean PR-AUC	T
A	15%	0.01622929	0.07216669	0.31500	0.75
A	30%	0.01623889	0.07219660	0.31395	0.75
A	100%	0.01622724	0.07215140	0.31465	0.75
B	15%	0.02290321	0.10789135	0.04736	1
B	30%	0.02290187	0.10789162	0.04723	1
B	100%	0.02288430	0.10782866	0.04870	1

Table 7. GBT results on each mode's eligible 2024 test set, equally averaged over the seven original thresholds. The 15% A row is from the teammate summary; the 15% B row is recomputed from a supplied per-threshold text export. Neither 15% pipeline was supplied. The 30% and 100% rows are recomputed from metric exports and reconciled against daily squared errors. Mean PR-AUC averages events with different prevalence and is supplemented by per-threshold results in the repository.

Increasing GBT training from 15% to 100% reduces mean Brier by approximately 0.013% for A and 0.083% for B. At 30%, A is slightly worse than at 15%, while B changes little. These are small descriptive differences, not established significance results. Each fraction undergoes depth and temperature selection, so this compares selected systems rather than changing sample size under a fixed fitting protocol. The saved 15% RF still has lower mean Brier than these GBT configurations in both information settings.

7.2 Training the original classifiers on all eligible examples

A separate full-training run repeats the original RF and logistic-regression workflow with a training fraction of 1.0. It retains the original year splits, features and seven evaluated thresholds, and compares RF depths 5 and 8 on validation. Both modes select RF depth 8 and temperature 1. These model packages are retained separately from the original 15% models.

Mode	Model	15% mean Brier	100% mean Brier	Change (%)
A	RF	0.01613051	0.01613638	+0.036
A	Logistic regression	0.01944150	0.01944300	+0.008
A	Historical frequency	0.02269628	0.02269628	+0.000
B	RF	0.02274285	0.02275362	+0.047
B	Logistic regression	0.02301006	0.02301076	+0.003
B	Historical frequency	0.02252051	0.02252051	+0.000

Table 8. Original 15% and supplied full-training results on each mode's eligible 2024 population. Positive change means higher error. Historical frequency uses the full eligible training population in both runs, so its unchanged score is a consistency check, not a new 15%-versus-100% fitting comparison.

The full-training RF and logistic-regression scores are close to their original values and do not show a consistent benefit from fitting on every eligible anchor. This agrees with the earlier validation study's finding that the reference configuration is relatively insensitive to increasing its fitting sample. Matching metadata, training amount supports, test counts and prevalence provide consistency evidence across exports; the imported archives do not contain row-level test predictions for an independent identity check. These follow-up 2024 results supplement, rather than replace, the original evaluation.

7.3 Thresholds inside the RF amount bins

The GBT experiments also evaluate 3, 7.5 and 15 mm, which are absent from the seven training query thresholds and lie inside the RF's amount bins. The archived job scores the original 15% RF and logistic baseline alongside GBT on the same loaded test frame. Table 9 reports the 100% GBT run; equivalent 30% results and individual thresholds are retained in the supporting CSVs.

Mode	q (mm)	RF Brier	GBT Brier	RF PR-AUC	GBT PR-AUC
A	3	0.010557	0.010707	0.2545	0.2534
A	7.5	0.005054	0.005120	0.1364	0.1388
A	15	0.001854	0.001913	0.0556	0.0557
B	3	0.012384	0.012857	0.0264	0.0273
B	7.5	0.005534	0.006141	0.0127	0.0122
B	15	0.001950	0.002620	0.0052	0.0042

Table 9. Additional 2024 query thresholds: original 15% RF versus 100% GBT. These compare fitted systems with different training fractions. RF has lower Brier in all six rows; PR-AUC differences vary by threshold.

These queries give direct evidence beyond the original threshold grid. The RF's empirical within-bin approximation remains competitive with threshold-conditioned GBT at these points, although three additional thresholds do not establish accuracy for every possible query. GBT does not impose monotonicity in q: increasing the requested rainfall amount can sometimes increase its predicted exceedance probability. Across all ten evaluated thresholds, adjacent-threshold violations occur in 0.0491% and 0.0681% of A's comparisons at 30% and 100% training, and in 0.5238% and 0.6164% for B. The main RF's survival-mixture construction is non-increasing by design.

These supplementary 2024 experiments leave the original models and test results unchanged. Runnable code, imported pipelines, source snapshots and reconciliation outputs accompany the report; the 15% A GBT row remains summary-only, while B has per-threshold metric evidence.

8 Discussion

The two tasks point to the same practical distinction. Calendar patterns explain when rain is more common across many years, whereas recent observations help distinguish the conditions surrounding a particular prediction time. The afternoon peak and seasonal variation therefore justify calendar features, but they do not imply that a calendar model can identify an individual storm. B's lack of overall improvement over historical frequency makes that limit visible. Model A has more useful short-term information, although its advantage narrows for large accumulations.

Missingness affects both conclusions. For exploration, coverage rules leave 85 of 96 year-months with at least one eligible station, while full-year summaries are supported in only three years. For prediction, A's history requirement retains 89.9% of B's eligible 2024 test cases. Performance on that subset does not establish what would happen during excluded periods. Neither a complete-window rule nor the modest sensitivity of monthly means to coverage cutoffs makes missing observations random.

There are also limits to reconstructing a forecast retrospectively. We assume that the observation timestamp marks the end of its five-minute accumulation interval. Requiring both update timestamps to be no later than the forecast time gives a conservative availability rule, but the annual files do not provide a complete archive of past revisions. For B, the saved evaluation uses coordinates attached to each historical observation, while the prediction interface uses a catalogue frozen from training. A station relocation could make those two representations differ. All 70 evaluated station IDs occur in the training catalogue, so performance at unseen stations is also untested. These issues matter when transferring the offline experiment into an operational service.

The follow-up narrows the uncertainty around our sampling choice. At the reference forest settings, doubling the sample gives little improvement in mean validation Brier. The deeper A forest does benefit slightly from 30%, so the result supports a practical compromise for the original configuration rather than a universal claim that more data are unnecessary. A controlled comparison fitting A and B to exactly the same station-time examples would still be needed to isolate the contribution of rainfall history. The supplementary queries at 3, 7.5 and 15 mm provide evidence inside the amount bins: the original RF has lower Brier than the full-training GBT at these points in both modes. This supports the tested queries without establishing the global within-bin approximation at every threshold.

The final-test uncertainty estimates are conditional on fitted models and one test year. Rolling-year evaluation would show whether the findings persist across different weather and station conditions; changing the time-block length would test the sensitivity of the reported interval. Controlled feature ablations and rolling-year evaluation remain future work; the additional threshold checks are reported in Section 7. The RF sampling and one-factor tuning study is reported on 2022 validation. The teammate GBT and full-training experiments additionally report supplementary 2024 results; both follow-ups were conducted after the original evaluation. It does not replace the original final-test evaluation. We therefore interpret the results as evidence for the observed network in 2024, with the strongest support for lower-threshold short-term prediction when recent station observations are available.

9 Conclusion

The records show clear seasonal, diurnal and local variation, but their gaps prevent a uniform account of annual rainfall across all eight years. Coverage-aware monthly summaries and common-station comparisons preserve useful evidence without treating missing periods as dry weather. For prediction, the recent-rainfall forest improves substantially on the calendar forest and the within-mode baselines in mean probability error. The remaining difficulty is most evident for rare, large accumulations. A usable probability estimate therefore needs both an appropriate model and a clear statement of the observations, threshold range and population to which its evaluation applies.

AI usage declaration

OpenAI Codex assisted with code development, data-cleaning diagnostics, cloud execution, feature and model implementation, tests, figure preparation, numerical analysis and report drafting. It also assisted with restructuring and editing this report and redrawing the calibration and December station figures from saved results. Reported numerical results come from the recorded data analyses and model outputs; synthetic records were used only for software tests. Group members remain responsible for checking the code, figures, interpretations and references in the submitted work.

References

[1] National Environment Agency. *Historical Rainfall across Singapore*, annual datasets 2017–2024. Data.gov.sg. https://data.gov.sg/collections/2279/view. Accessed 9 October 2026.

[2] Meteorological Service Singapore. *Climate of Singapore*. https://www.weather.gov.sg/climate-climate-of-singapore/. Accessed 9 October 2026.

[3] geoBoundaries / Urban Redevelopment Authority. Singapore outline and planning-area geometries. Source and licence records are retained in the project assets; planning-area geometry is based on URA Master Plan 2014.

[4] Apache Spark. *Classification and regression*, Spark 3.5.3 documentation. https://dlcdn.apache.org/spark/docs/3.5.3/ml-classification-regression.html. Accessed 9 October 2026.

[5] Apache Spark. *BinaryClassificationMetrics*, Spark 3.5.3 Java API. https://spark.apache.org/docs/3.5.3/api/java/org/apache/spark/mllib/evaluation/BinaryClassificationMetrics.html. Accessed 9 October 2026.

A Distribution and source-gap evidence

These supplementary figures separate the distribution of measured rain from the absence of measurements. They support the data-quality discussion without treating an unrecorded interval as dry weather.

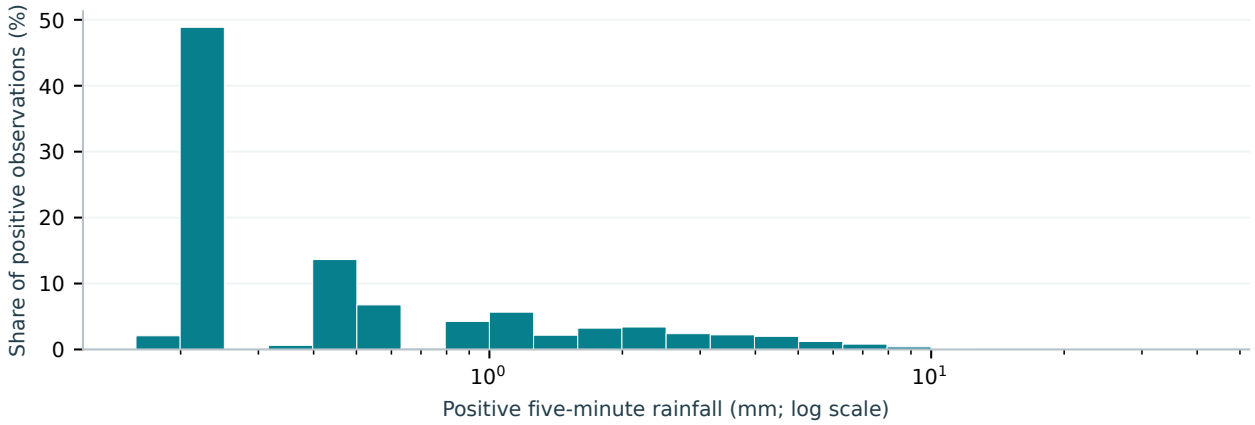


Figure A.1. Distribution of positive five-minute rainfall across 2017–2024. Bars give the share of positive observations in log-spaced bins; the horizontal axis is logarithmic. Recorded zeros are excluded. The long upper tail motivates evaluating rare accumulations separately.

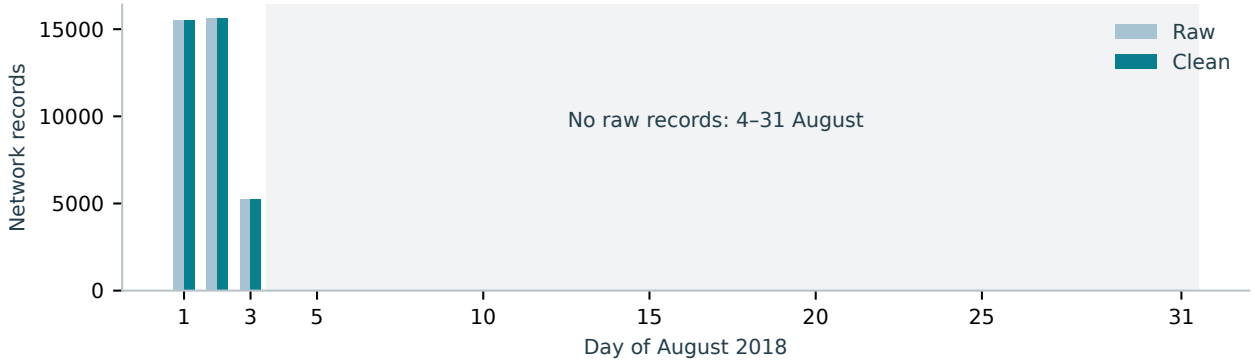


Figure A.2. Daily network-wide raw and cleaned record counts in August 2018. The absence of observations on 4–31 August is already present in the source file. The cleaning process therefore does not explain this gap, and this figure does not identify its upstream cause.

B Monthly trajectories and annual support

The monthly trajectories retain usable periods individually. The annual plot applies the stricter full-year rule, showing why monthly evidence is more extensive than annual evidence.

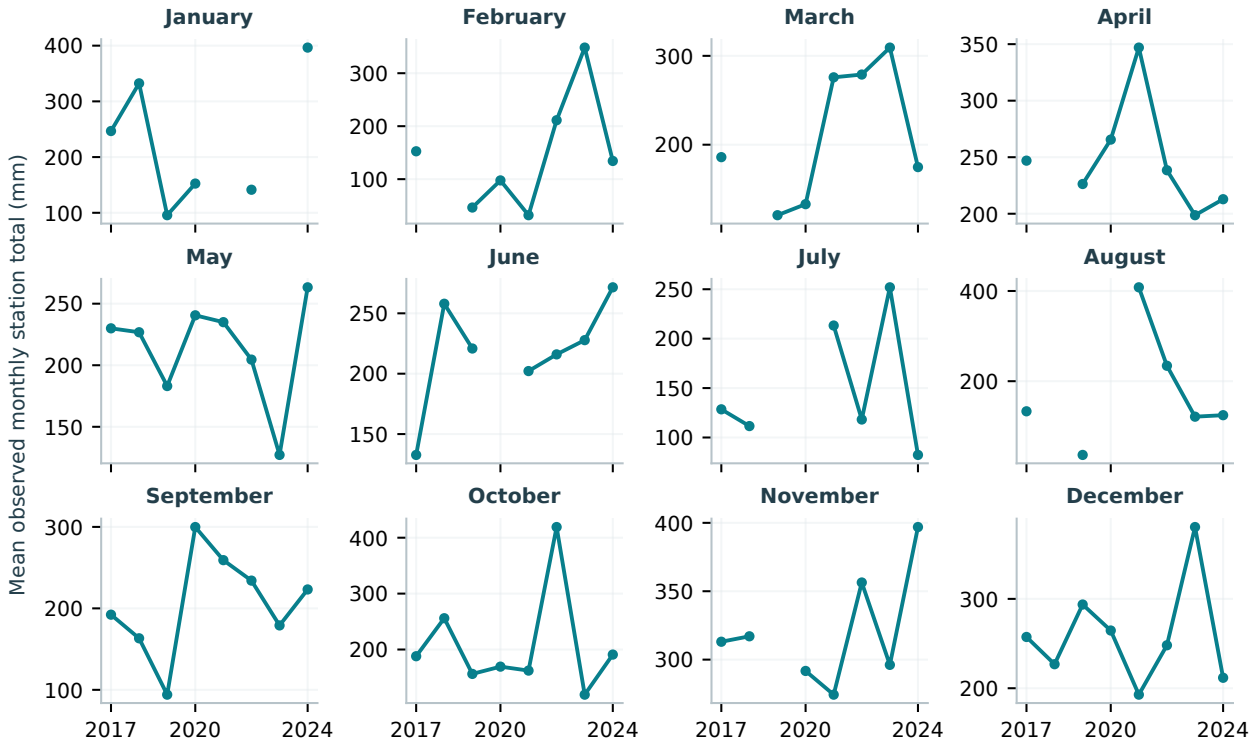


Figure B.1. Mean observed monthly station totals, with a separate trajectory for each calendar month. Missing year-months break the lines. Vertical ranges differ across panels; use the main monthly heatmap to compare absolute amounts between months. Station composition can change between points, so these are descriptive network trajectories.

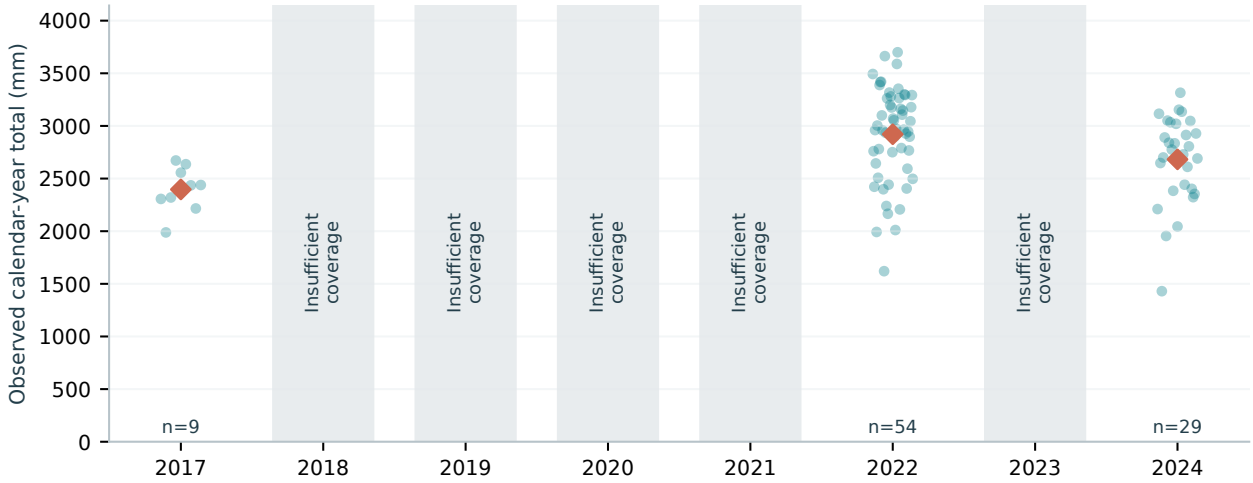


Figure B.2. Observed calendar-year totals at eligible stations. Points show station totals, coral diamonds their mean, and n the contributing station count. Grey columns have no station meeting annual coverage of at least 95% and coverage of at least 90% in every month. The three supported years use different station sets and do not establish a national wettest-year ranking.

C Daily rainfall and coverage sensitivity

Daily upper-tail amounts and wet-day frequencies describe different aspects of rainfall. The sensitivity curves show how changing the completeness rule affects seasonal summaries, while retaining the limitations of the observed network.

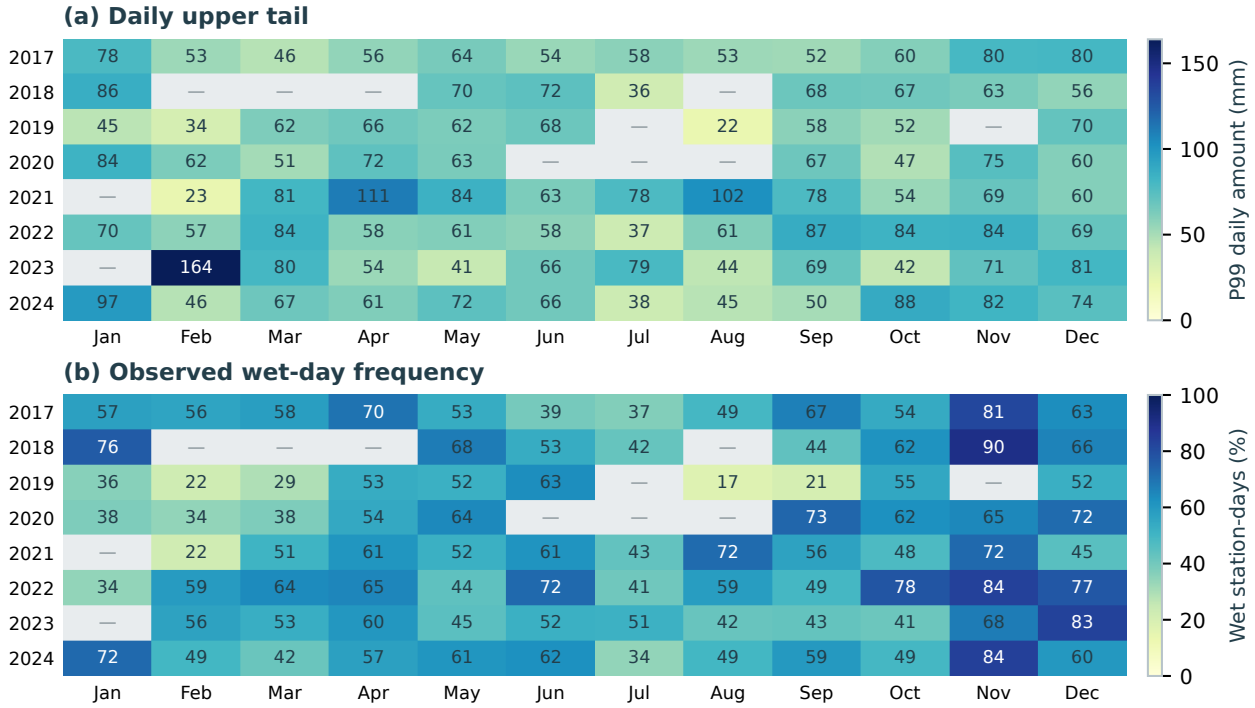


Figure C.1. Monthly daily-rainfall P99 (upper) and wet station-day percentage (lower). Included days have at least 95% daily coverage within eligible station-months. Quantiles include dry days; wet means observed daily rainfall above zero. P99 is an approximate Spark quantile of station-days, not an independent-storm return level. Grey cells lack qualifying support.

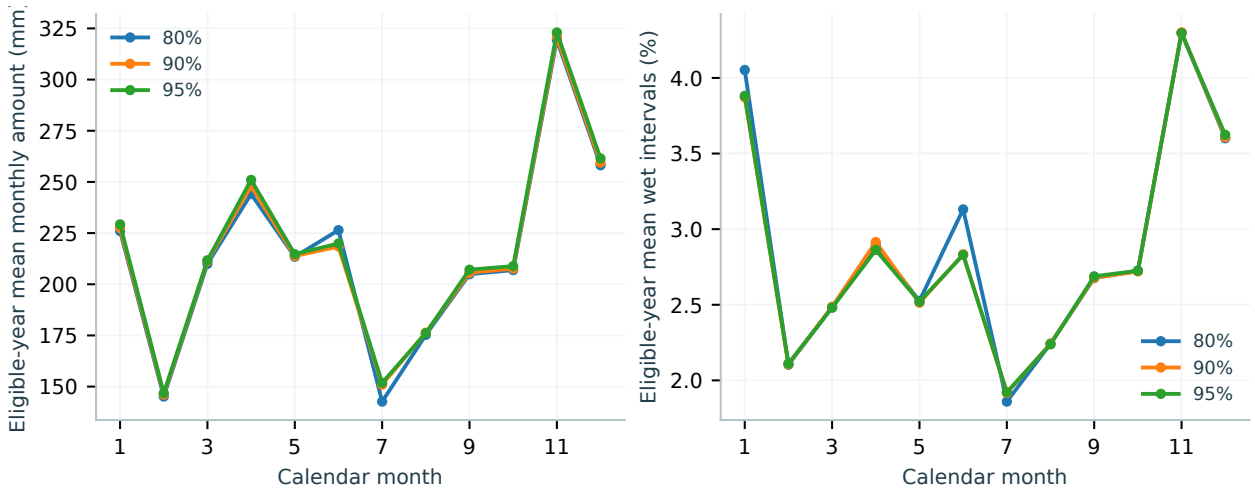


Figure C.2. Seasonal mean amount and wet-interval frequency under 80%, 90% and 95% monthly coverage cutoffs. Each curve uses its own eligible stations and equally weighted available years. Similar curves support the stability of these summaries under the tested cutoffs; they do not establish random missingness.

D Additional spatial views

The twelve-month view complements the main December comparison. Regional colours summarise contributing gauges and should be read together with their locations and coverage.



Figure D.1. All twelve months of 2024. Colours show observed monthly totals at qualified stations, and n gives their number. The original gallery's global monthly colour scale is retained; it differs from the December-only scale in the main text. The eligible network changes by month, and no rainfall is interpolated between gauges.

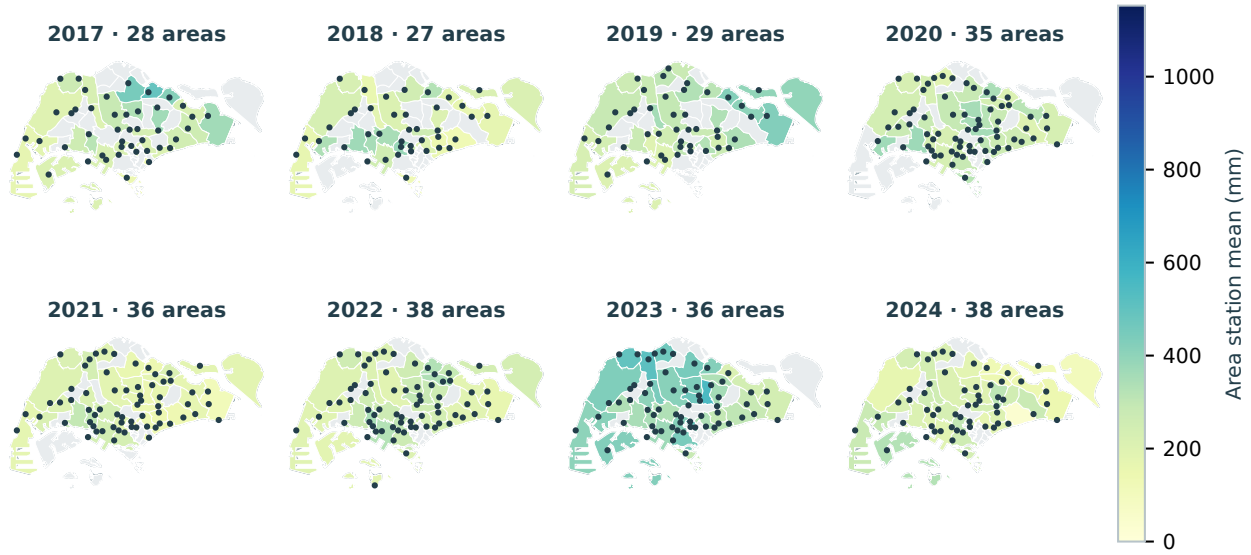


Figure D.2. December planning-area station means across eight years. Each coloured polygon shows the arithmetic mean of its assigned eligible stations; dots show contributing locations. Grey means no supporting station, not zero rainfall. The original regional scale is retained. These are gauge summaries, not area-weighted rainfall estimates; station counts and placements affect their representativeness.

E Additional predictive diagnostics

These figures retain the threshold-specific baseline comparison and diagnostics for the original evaluated models. They do not evaluate the later sensitivity candidates on the final test set.

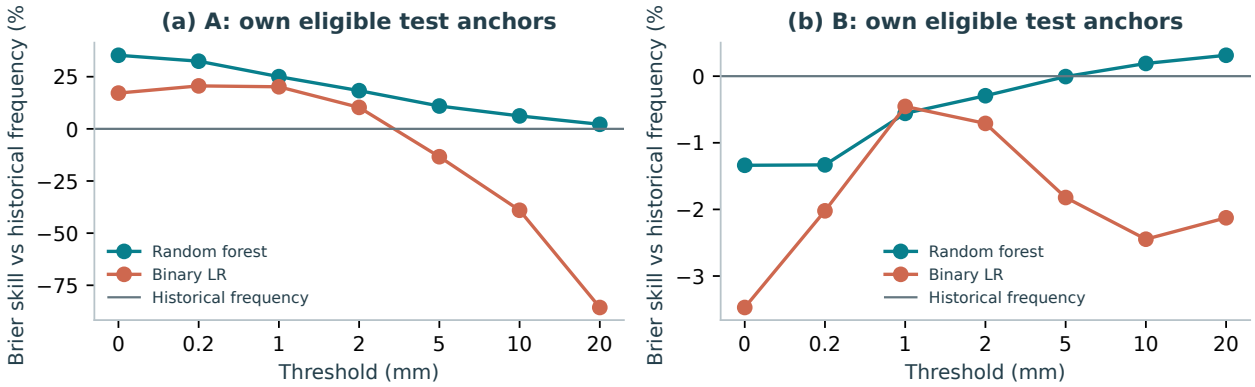


Figure E.1. Brier skill relative to each mode's historical-frequency baseline on its own eligible 2024 test population. Positive values indicate lower error than the baseline; zero indicates equal error. A's random forest improves on the baseline at every evaluated threshold, whereas A's logistic regression deteriorates at larger thresholds. B's forest has small, threshold-dependent gains and losses. The two panels use different eligible populations.

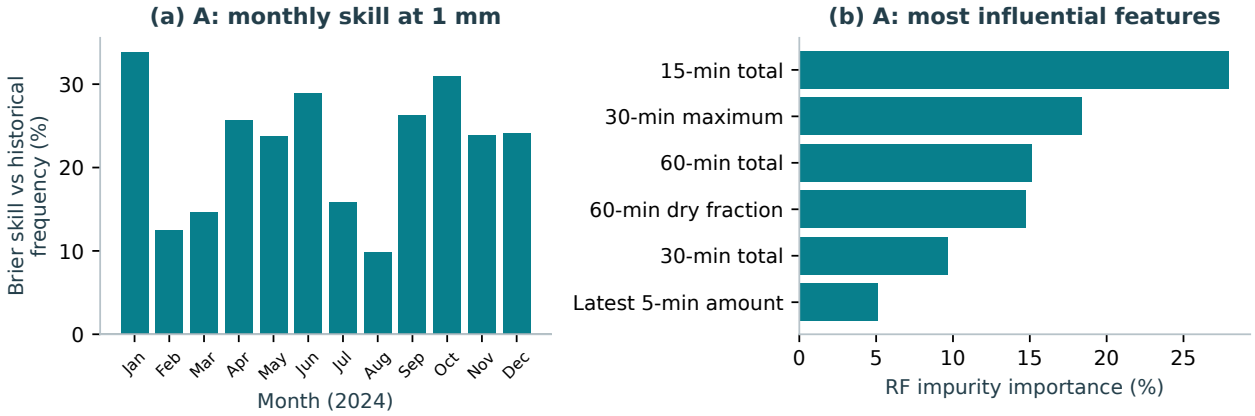


Figure E.2. Left: A's monthly Brier skill relative to historical frequency at a rainfall threshold of 1 mm in 2024. Right: its six largest random-forest impurity importances. Positive monthly skill describes this holdout. Correlated rainfall summaries share information, so importance does not identify an independent feature effect or establish causation.